



ORACLE

MySQL Implementation Essentials Bootcamp

Introduction

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2023



Course Intro

Welcome!

Timings and “Logistics”

Day 1: Introduction
MySQL Enterprise
Oracle Cloud Infrastructure overview
Installation and Architecture - part1

Day 2: Installation and Architecture – part 2
Database Design

Day 3: MySQL Security
MySQL Backup – part 1

Day 4: MySQL Backup – part 2
MySQL Replication
High Availability

Day 5: MySQL Enterprise Monitor
MySQL HeatWave Database Service



- **Start**
9:00 AM
- **Break ~ 11:00**
- **End**
1:00 PM

- **Please**

Mute microphone when not needed



Be in time

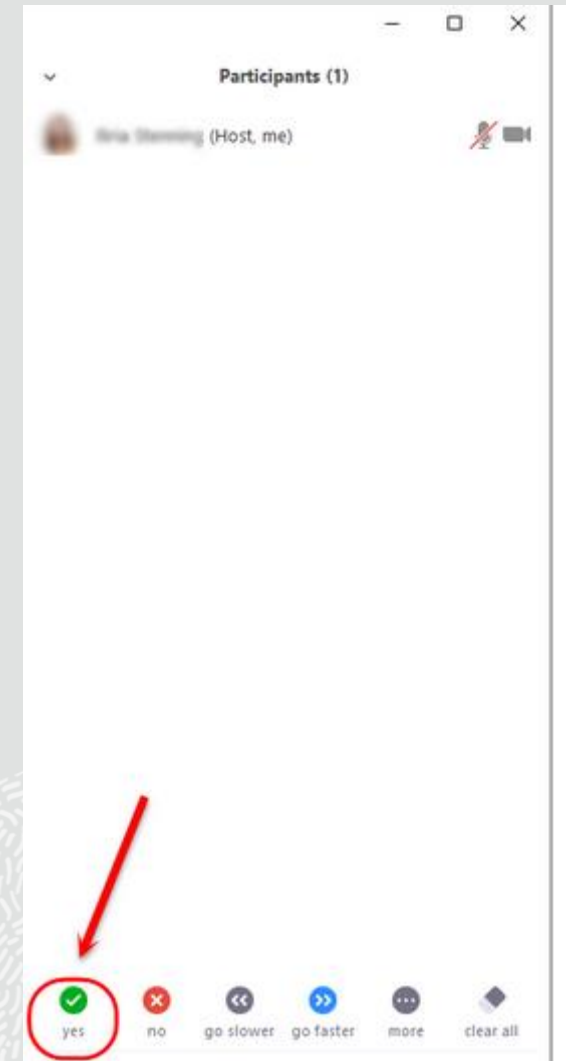
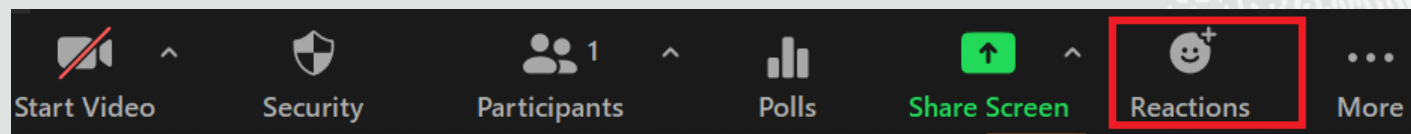
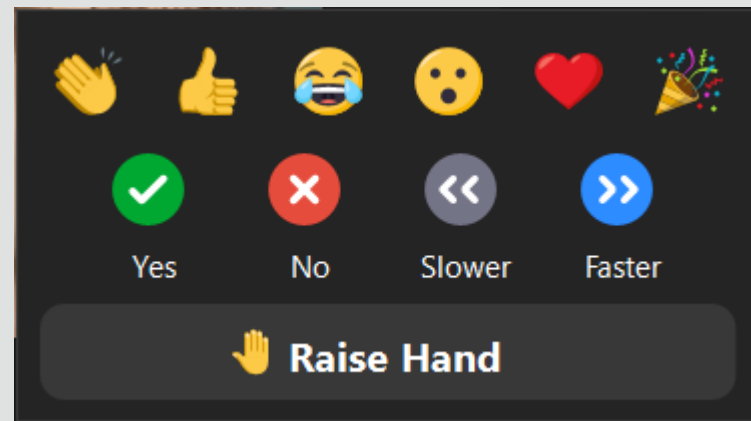


Use coffee icon during breaks



Guidelines

Use Zoom reactions to answer to questions
Use chat or verbally ask questions



Introduce yourself

- Your Name (how to pronounce?)
- Your role: are you a DBA, a developer, an architect, or...?
- Do you know MySQL?
 - Do you know MySQL Enterprise?
 - Which version do you know?
- Do you know Oracle Cloud Infrastructure?
- Why are you interested in this workshop on MySQL?



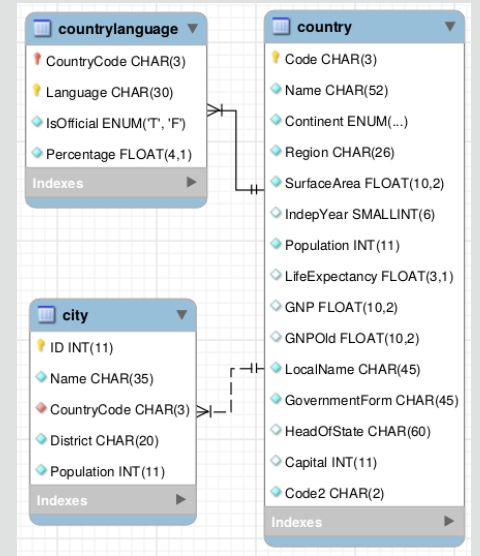
Program Agenda

- **Introduction**
- Oracle MySQL Enterprise
- Oracle Cloud Infrastructure overview
- Installation and Architecture
- Database Design
- Security
- Backup
- Replication
- High Availability
- MySQL Enterprise Monitor
- HeatWave

Introduction

What is MySQL?

- Relational Database
 - Transactional, ACID
 - InnoDB storage engine: ARIES, MVCC
 - OLTP: low latency, high throughput
- Replication
 - Read scale out, High Availability/Cluster
- Simple, Solid, Secure
 - Easy to use, Proven at scale



like Microsoft SQL , DB2, Sybase, Postgres ... 😊

MySQL: Ubiquity & Market Share

402 systems in ranking, January 2023

Rank			DBMS	Database Model	Score		
Jan 2023	Dec 2022	Jan 2022			Jan 2023	Dec 2022	Jan 2022
1.	1.	1.	Oracle +	Relational, Multi-model i	1245.17	-5.14	-21.72
2.	2.	2.	MySQL +	Relational, Multi-model i	1211.96	+12.56	+5.91
3.	3.	3.	Microsoft SQL Server +	Relational, Multi-model i	919.39	-4.96	-25.43
4.	4.	4.	PostgreSQL +	Relational, Multi-model i	614.85	-3.13	+8.29
5.	5.	5.	MongoDB +	Document, Multi-model i	455.18	-14.15	-33.38

Source: <https://db-engines.com/en/ranking>

#1 Open Source Database
#2 Most Popular Database

What's in it for partners?

- Leading database for Web (LAMP) and new standards (e.g. OpenStack)
- New markets, scale out quickly
- Great choice for projects where end customer TCO is critical
- High margins, high growth potential
- Ongoing subscription revenue, keep partners engaged long term
- Premier lifetime support, high customer satisfaction
- Will work alongside Oracle DB or SQL Server with projects
- SQL Server and Oracle dba's can be easily trained
- Flexible deployment options: on-prem, OCI (mngd service), AWS/Azure, BYOL, etc...

MySQL Expertise Program



MySQL Service Expertise :

- Recognizes partners with proficiency to implement MySQL Enterprise Edition
- Expertise Logo
- Priority placement in Oracle Partner Finder
- Free Support services requests
- Social Promotion of Joint Customer Successes
- Access to Demonstration Services
- Eligible for MDF

Bootcamp

- Focus on MySQL 8.0 Enterprise
- Course structured in 13 Units
 - Template for OCS Exam
 - Exam cover MySQL 8.0
- Each unit has hands-on exercises
- Focus on course is based on operations, less DDL and DML
- Slides and labs are available for download
 - follow trainer instructions

Hands-on exercise

- Labs are completed by each student
- Each student works with 4 Linux servers and one MDS
- Server are hosted inside Oracle Public Cloud
 - Accessible with SSH protocol with key (no username/password)
- **Please, all passwords are/have to be set to "Welcome1!"**
 - if you change them, it's your responsibility remember... 
 - Some security labs require to test a different password, there is a meaning to use exactly what proposed ;)

Requirements

For lectures and labs

- **Required Skills**
 - Basic knowledge of Linux
 - Basic knowledge of SQL & databases concepts
- **Required HW**
 - Laptop/Notebook
- **Required SW**
 - SSH client
 - (optional) VNC/RDP client

To receive free OU exam attempt*

- Pass the [MySQL Solutions Engineer Specialist](#) Guided Learning Path
 - free online course incl. Oracle Specialist Assessment
- Pass the [MySQL HeatWave Solutions Engineer Specialist](#) Guided Learning Path
 - free online course incl. Oracle Specialist Assessment
- Attend all Bootcamp days
- Contact your local MySQL A&C rep to request

**Pending availability*

Exam 1Z0-921 - MySQL 2021 Implementation Essentials

- Where take the exam?
 - Oracle University
- Questions
 - 52 questions (Multiple Choice)
 - 60% passing score (-> 32 question to pass, or max 20 wrong)
 - 90 minutes
- Tips
 - Take your time to study !
 - Don't be anxious
 - Use time wisely
 - Details: https://education.oracle.com/mysql-2021-implementation-essentials/pexam_1Z0-921



Oracle Certification Program / Trainings

Courses

- MySQL Explorer (free)
- MySQL Fundamentals
- MySQL 8.0 for Database Administrators
- MySQL 8.0 for Developer
- MySQL Performance Tuning
- And many others

Certifications

- MySQL 2021 Implementation Essentials (*)
- MySQL 8.0 Database Administrator
- MySQL 8.0 Database Developer

https://education.oracle.com/database/mysql/pFamily_406

Latest Updates...

MySQL within Oracle

More Investment, More Innovation



... More than 25 Years MySQL (1995)

...InnoDB Oracle Stewardship since 2005

...MySQL within Oracle since 2010



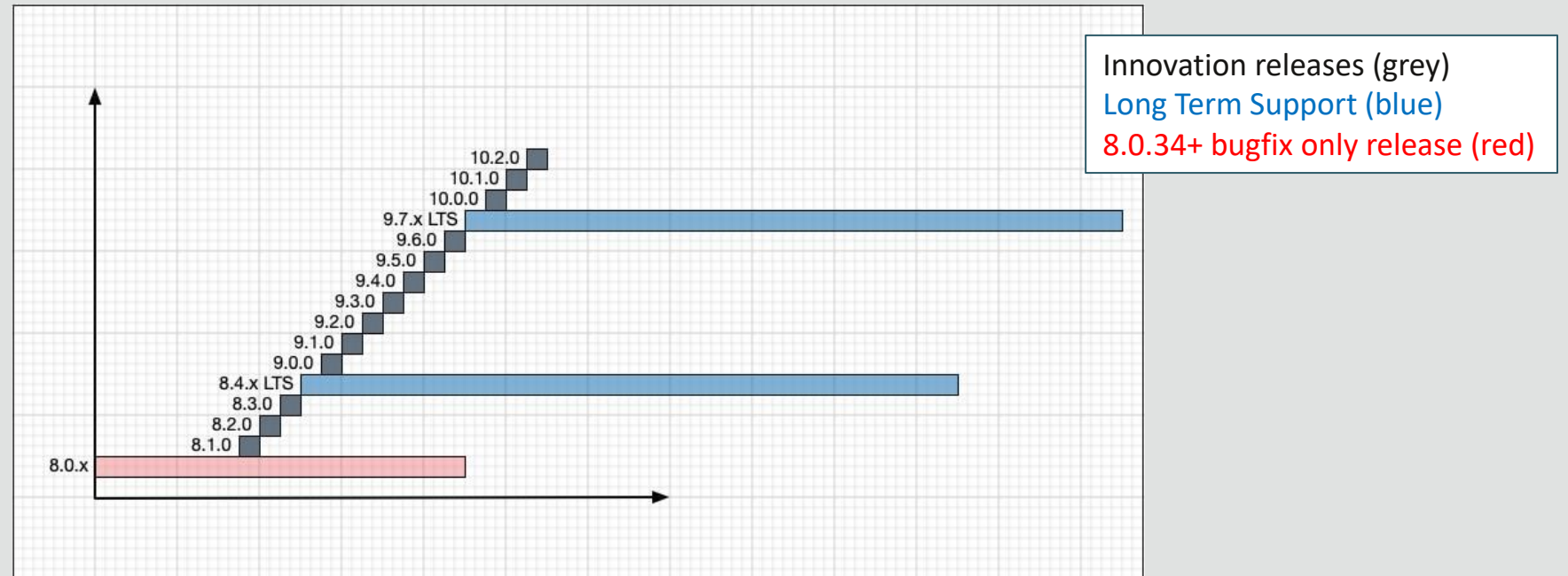
MySQL Status (So far...)



- **Oracle Premier Support: 5 Years after General Availability**
 - Release MySQL 5.5: Dec. 2010 → Dec. 2015
 - Release MySQL 5.6: Feb. 2013 → Feb. 2018
 - Release MySQL 5.7: Oct. 2015 → Oct. 2020
 - Release MySQL 8.0: Apr. 2018 → Apr. 2025
- **Oracle Extended Support: 3 Years after Premier Support (8 Years after GA)**
 - Release MySQL 5.5: Dec. 2015 → Dec. 2018
 - Release MySQL 5.6: Feb. 2018 → Feb. 2021
 - Release MySQL 5.7: Oct. 2020 → Oct. 2023
 - Release MySQL 8.0: Apr. 2025 → Apr. 2026
- **Oracle Sustaining Support**
 - Lifetime



NEW: MySQL LTS & Innovation Releases



MySQL Long-Term Support (LTS) Releases

- stable: bugfix & security patches only
- backwards compatibility
- will likely released every 2 years
- support lifecycle: 5y premier + 3y extended

MySQL Innovation Releases

- leading-edge innovations
- easy migration between LTS & Innovation
- will likely released every quarter
- support lifecycle: short term

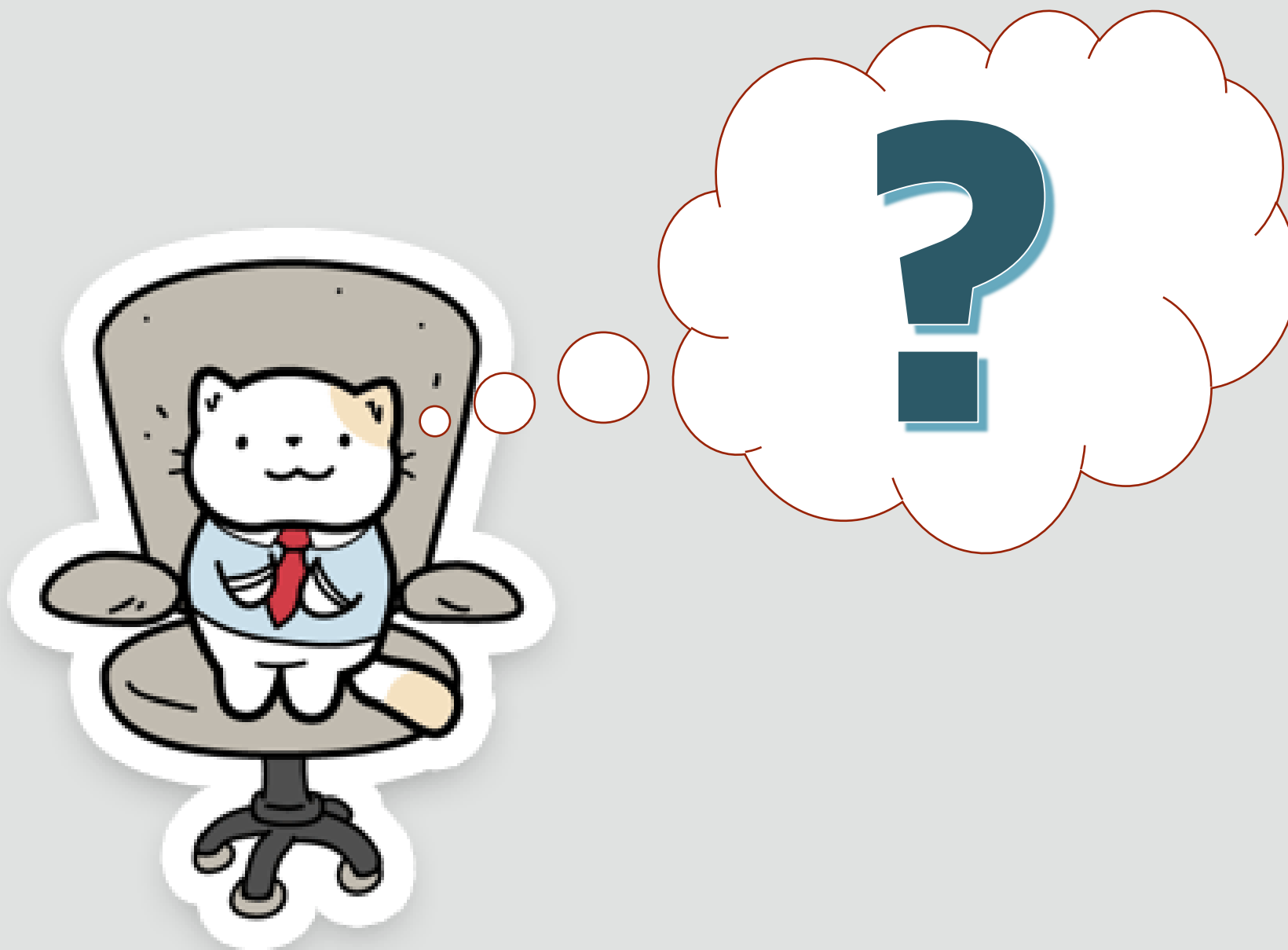
MySQL LTS & Innovation Releases (contd.)

- LTS and Innovation releases will be available for Community, Enterprise and MySQL HeatWave Database Service
- LTS and Innovation releases will be Generally Available and are recommended to be used in production environments
- LTS and Innovation releases are both commercially supported
- MySQL tools will follow same approach
- Connectors will use the latest version used, but will be considered Generally Available and will remain compatible with all supported MySQL Server versions
- Details here
 - <https://blogs.oracle.com/mysql/post/introducing-mysql-innovation-and-longterm-support-lts-versions>

MySQL– Key updates

- MySQL 8.0 with more than 250 new features
 - Data Dictionary/Atomic DDL
 - Security Roles/ SHA2 password plugin
 - MySQL Document Store (X Dev API) & MySQL Connector Node.js / MySQL Shell
 - And much more
 - <https://dev.mysql.com/doc/refman/8.0/en/mysql-nutshell.html>
- MySQL Enterprise Backup 8.x*
- MySQL Enterprise Monitor 8.x*
- MySQL Cluster 8.0 [NDB Storage engine]
- and new features every new release

<https://mysqlserverteam.com/the-complete-list-of-new-features-in-mysql-8-0/>





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